

EV-Drivetrain-Resonance-and-Control

Rigid- and Flexible-Shaft Modeling, Simulation, and Shaft Design

1 Purpose and modeling scope

The original mechatronic system consists of an electrical circuit with resistance R and inductance L , a DC motor with motor constant k , and a rotating load with polar moment of inertia I_p . The original model assumes a rigid shaft. The supplied parameters are

$$R = 1 \, \Omega, \quad L = 1 \, \text{H}, \quad I_p = 1 \, \text{kg m}^2, \quad k = 10.$$

The source voltage V_s and load moment M_L are the inputs. The original motor equations are

$$M_M = -ki_M, \quad v_M = k\omega_M.$$

The original mechanical sign convention is

$$\dot{\omega}_M = \frac{1}{I_p}(M_L - M_M).$$

Because $M_M = -ki_M$,

$$I_p\dot{\omega} = M_L + ki_M.$$

The flexible-shaft extension is designed in two stages:

1. simulate the rigid system and obtain its maximum occurring motor torque;
2. use that torque as the shaft-design load, calculate the shaft diameter and torsional stiffness, and then build the flexible model.

A separate motor rotor inertia is not supplied. Therefore no numerical J_M is introduced.

2 Rigid-shaft model

2.1 State variables, inputs, and outputs

For the rigid model,

$$\omega_R = \omega_M = \omega.$$

The state vector is

$$\mathbf{x}_r = \begin{bmatrix} i_M \\ \omega \end{bmatrix},$$

and the input vector is

$$\mathbf{u} = \begin{bmatrix} V_s \\ M_L \end{bmatrix}.$$

The main outputs are

$$\mathbf{y}_r = \begin{bmatrix} i_M \\ \omega \end{bmatrix}.$$

Additional report quantities are

$$v_R = Ri_M, \quad v_M = k\omega, \quad v_L = L\frac{di_M}{dt}, \quad T_M = ki_M.$$

2.2 Kirchhoff voltage law

Kirchhoff's voltage law gives

$$V_s - v_R - v_L - v_M = 0.$$

Substitution gives

$$L\dot{i}_M = V_s - Ri_M - k\omega,$$

or

$$\dot{i}_M = -\frac{R}{L}i_M - \frac{k}{L}\omega + \frac{1}{L}V_s.$$

The mechanical equation is

$$I_p\dot{\omega} = M_L + ki_M,$$

hence

$$\dot{\omega} = \frac{k}{I_p}i_M + \frac{1}{I_p}M_L.$$

2.3 Rigid state-space representation

$$\dot{\mathbf{x}}_r = A_r\mathbf{x}_r + B_r\mathbf{u}$$

with

$$A_r = \begin{bmatrix} -\frac{R}{L} & -\frac{k}{L} \\ \frac{k}{I_p} & 0 \end{bmatrix}, \quad B_r = \begin{bmatrix} \frac{1}{L} & 0 \\ 0 & \frac{1}{I_p} \end{bmatrix}.$$

For the supplied parameters,

$$A_r = \begin{bmatrix} -1 & -10 \\ 10 & 0 \end{bmatrix}.$$

The characteristic equation is

$$\lambda^2 + \lambda + 100 = 0,$$

giving

$$\lambda_{1,2} = -0.5 \pm j 9.9875.$$

Therefore

$$f_r \approx 1.5896 \text{ Hz}, \quad T_r \approx 0.6291 \text{ s}.$$

3 Maximum torque from the rigid model

The flexible shaft is sized from the maximum torque occurring in the rigid model. The required pulse periods are

$$\frac{T_r}{2}, \quad T_r, \quad 2T_r, \quad 10T_r.$$

For the resonant $P = T_r$ case,

$$T_M = ki_M.$$

The rigid simulation gives approximately

$$\max |i_M| \approx 0.63676 \text{ A},$$

and therefore

$$T_{\max,r} \approx 6.3676 \text{ N m}.$$

The MATLAB script `RigidShaft_MaxTorque.m` calculates this value directly and saves it to `RigidShaft_MaxTorque.mat`. The flexible parameter file reads this file, so the flexible shaft is explicitly sized from the rigid-body simulation.

4 Flexible-shaft design

4.1 Design choices for the added shaft

The original problem does not provide the material, length, allowable stress, or safety factor for the new shaft. These are therefore explicit design choices for the added component:

$$G = 79 \text{ GPa}, \quad l_s = 0.30 \text{ m}, \quad \tau_{\text{allow}} = 50 \text{ MPa}, \quad S_F = 2.$$

The design torque is

$$T_{\text{design}} = S_F T_{\text{max},r} \approx 12.7352 \text{ N m}.$$

For a solid circular shaft,

$$\tau_{\text{max}} = \frac{16T}{\pi d^3}.$$

Thus,

$$d_{\text{min}} = \left(\frac{16T_{\text{design}}}{\pi \tau_{\text{allow}}} \right)^{1/3}.$$

Numerically,

$$d_{\text{min}} \approx 10.906 \text{ mm}.$$

The selected preferred diameter is

$$d = 12 \text{ mm}.$$

4.2 Shaft torsional stiffness

The polar second moment is

$$J_p = \frac{\pi d^4}{32} \approx 2.035752e-09 \text{ m}^4.$$

The torsional stiffness is

$$K_s = \frac{GJ_p}{l_s}$$

and the selected shaft gives

$$K_s \approx 536.081 \text{ N m/rad}.$$

5 Flexible-shaft model without an assumed motor inertia

A classical two-inertia model would require a motor rotor inertia J_M . Because the original problem does not provide it, this report does not invent a value.

Instead, the original neglect of motor-side inertia is retained. Therefore the motor torque and shaft torque are in instantaneous equilibrium:

$$T_s = T_M = k i_M.$$

For the elastic shaft,

$$T_s = K_s(\theta_M - \theta_R).$$

Therefore

$$\theta_M - \theta_R = \frac{k}{K_s} i_M.$$

Differentiating,

$$\omega_M - \omega_R = \frac{k}{K_s} \dot{i}_M,$$

so

$$\omega_M = \omega_R + \frac{k}{K_s} \dot{i}_M.$$

The electrical equation is

$$L \dot{i}_M = V_s - R i_M - k \omega_M.$$

Substituting the flexible-shaft kinematics gives

$$L \dot{i}_M = V_s - R i_M - k \omega_R - \frac{k^2}{K_s} \dot{i}_M.$$

Hence

$$\left(L + \frac{k^2}{K_s} \right) \dot{i}_M = V_s - R i_M - k \omega_R.$$

Define

$$L_{\text{eff}} = L + \frac{k^2}{K_s}.$$

For the designed shaft,

$$L_{\text{eff}} \approx 1.186\,539\,H.$$

The flexible current state equation is

$$\dot{i}_M = -\frac{R}{L_{\text{eff}}} i_M - \frac{k}{L_{\text{eff}}} \omega_R + \frac{1}{L_{\text{eff}}} V_s.$$

The roll equation remains

$$\dot{\omega}_R = \frac{k}{I_p} i_M + \frac{1}{I_p} M_L.$$

5.1 Flexible state variables, inputs, and outputs

The independent states are

$$\mathbf{x}_f = \begin{bmatrix} i_M \\ \omega_R \end{bmatrix}.$$

The inputs are

$$\mathbf{u} = \begin{bmatrix} V_s \\ M_L \end{bmatrix}.$$

Useful outputs are

$$\mathbf{y}_f = \begin{bmatrix} i_M \\ \omega_R \\ \omega_M \\ T_s \\ \Delta\theta \end{bmatrix},$$

where

$$T_s = k i_M, \quad \Delta\theta = \frac{k}{K_s} i_M, \quad \omega_M = \omega_R + \frac{k}{K_s} \dot{i}_M.$$

5.2 Flexible state-space representation

$$\dot{\mathbf{x}}_f = A_f \mathbf{x}_f + B_f \mathbf{u}$$

with

$$A_f = \begin{bmatrix} -\frac{R}{L_{\text{eff}}} & -\frac{k}{L_{\text{eff}}} \\ \frac{k}{I_p} & 0 \end{bmatrix},$$

and

$$B_f = \begin{bmatrix} \frac{1}{L_{\text{eff}}} & 0 \\ 0 & \frac{1}{I_p} \end{bmatrix}.$$

The output matrices for

$$\mathbf{y}_f = [i_M, \omega_R, \omega_M, T_s, \Delta\theta]^T$$

are

$$C_f = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{kR}{K_s L_{\text{eff}}} & 1 - \frac{k^2}{K_s L_{\text{eff}}} \\ \frac{k}{K_s} & 0 \\ \frac{k}{K_s} & 0 \end{bmatrix},$$

and

$$D_f = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{k}{K_s L_{\text{eff}}} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The designed flexible model has

$$f_f \approx 1.4596 \text{ Hz}, \quad T_f \approx 0.6851 \text{ s}.$$

6 File structure and execution order

The project uses separate rigid and flexible files:

```
RigidShaft_parameters.m
build_RigidShaft_Simulink.m
RigidShaft_MaxTorque.m
RigidShaft_MaxTorque.mat % generated

FlexibleShaft_parameters.m
build_FlexibleShaft_Simulink.m

EV_Drivetrain_Rigid_and_Flexible_Report.tex
```

Run:

```
RigidShaft_parameters
build_RigidShaft_Simulink

RigidShaft_MaxTorque

FlexibleShaft_parameters
build_FlexibleShaft_Simulink
```

The builder scripts create:

```
EV_RigidShaft.slx
EV_FlexibleShaft.slx
```

7 Conclusion

The rigid system is evaluated first using only the supplied plant parameters. Its maximum motor torque is then transferred directly into the shaft design. The calculated rigid-model maximum is approximately

$$T_{\max,r} \approx 6.3676 \text{ N m.}$$

With the documented shaft-design choices,

$$d_{\min} \approx 10.906 \text{ mm,}$$

so a 12 mm shaft is selected and

$$K_s \approx 536.081 \text{ N m/rad.}$$

The flexible model introduces nonzero shaft twist and allows $\omega_M \neq \omega_R$ during transients, while avoiding an unsupported numerical motor inertia.